software project

Campus Lost and Found System

Problem: Students frequently lose items like water bottles, chargers, books, and ID cards around campus. Currently, there is no centralized way to report lost items or claim found items. Students post on social media groups, but posts get lost quickly, and there is no verification system. This leads to many items never being returned to their owners.

Solution: Build a web application where students can report lost items with photos, descriptions, and last known locations. Other students can report found items with similar details. The system will use simple matching algorithms to suggest possible matches between lost and found reports. Users can claim items by providing proof of ownership through photos or unique identifiers. The system includes a rating feature to build trust in the community. Notifications are sent when potential matches are found. Admin users can verify claims before releasing items. This project teaches database design, user authentication, image upload handling, search algorithms, and notification systems.

week4 | 14 _ 17 SEP :

▼ System Scenario

> Ahmed is a university student who attends a lecture in the Engineering Building. After the lecture ends, he goes to the library and later realizes th
> ”?.;NMB NVt he has lost his black laptop charger. Ahmed is not sure exactly where he left it, but he remembers using it in the Engineering Building during the morning.

Ahmed opens the **Campus Lost and Found System** and logs into his account. He selects **Report Lost Item** and creates a new report.

Ahmed enters the item name as “Laptop Charger,” selects the category “Electronics,” enters the color “Black,” adds the charger brand, and writes a short description. He also selects the Engineering Building as the last known location and enters the date and approximate time when he last saw the charger. In addition, he uploads an old photo of the charger to help identify it.

After Ahmed submits the report, the system stores the information and checks existing found-item reports for possible matches. At that time, there is no strong match, so the system confirms that the lost-item report has been created successfully.

Ahmed can later return to the system to view the report, edit some details if needed, manually search found items, and check whether any possible matches have appeared.

Later that same day, another student named Khalid finds a black laptop charger near a desk in the Engineering Building. Khalid opens the Campus Lost and Found System and logs into his account. He selects **Report Found Item** and enters the charger details.

Khalid selects the category “Electronics,” enters the color “Black,” adds the brand if it is visible, writes that the charger was found near a desk in the Engineering Building, and enters the date and time when it was found. He also uploads a photo of the charger.

After submitting the report, Khalid hands the charger to the Lost and Found **Administrator**. The system stores the found-item information and compares it with existing lost-item reports. The system notices that Ahmed’s report has several matching details, such as category, color, building, date, and description. Because of these similarities, the system identifies Ahmed’s report as a possible match.

The system then sends Ahmed a match notification informing him that a found item may be the charger he lost.

Ahmed opens the notification and views the found-item details. He sees the photo and description and believes that the charger may be his.

Ahmed selects **Submit Claim** and submits the claim directly. The system records the claim and automatically changes its status to **Pending**. Ahmed can view the

claim status from his account while waiting for a decision.

An **Administrator** logs into the system and opens the list of pending claims. The administrator selects Ahmed's claim and reviews the lost-item report, the found-item report, and the information related to the claim.

The administrator compares the information, such as the description, location, date, photo, model, and identifying marks, to determine whether Ahmed is the real owner of the charger.

If the information is consistent, the administrator approves the claim. The system changes the claim status from **Pending** to **Approved** and sends Ahmed a notification informing him that his claim has been accepted.

The administrator then arranges for the charger to be returned to Ahmed through the university Lost and Found office.

After Ahmed receives the charger, the administrator changes the item status to **Returned**, and the related lost and found reports are marked as **Resolved**.

If the information does not match sufficiently, the administrator can reject the claim. In that case, the system changes the claim status to **Rejected** and sends Ahmed a notification informing him that the claim was not approved.

The system also allows Ahmed to manually search through reported found items even before receiving a suggested match. For example, he can filter search results by category, location, date, or keyword. He can open an item to view its details and, if he believes it belongs to him, submit a claim.

The system also allows users who find items to report them even when there is no existing lost-item report. These reports remain stored in the system and can later be matched with a new lost-item report if another student reports a similar missing item.

Administrators can also manage reports in the system. They can review reports that appear incorrect, duplicated, inappropriate, or suspicious and take the necessary action. They can also monitor unresolved items and update their status when an item is returned to its owner.

The system therefore supports the complete process from the moment an item is lost or found until it is successfully returned to its owner or remains available in the system for future matching.

Main flow:

Student loses item → reports lost item → another student finds item → reports found item → hands item to administrator → system compares reports → possible match is suggested → lost-item owner receives notification → submits claim → claim becomes Pending → administrator reviews and verifies → claim is approved or rejected → item is returned → case is marked Resolved.

▼ Actors

i- Lost Item Owner / Student

- reports a lost item
- searches items
- views suggested matches
- submits claims
- receives notifications
- checks claim status

ii- Finder / Student

- reports a found item
- provides item details/photo

iii- Administrator

- reviews claims
- verifies ownership
- manages reports
- marks returned items

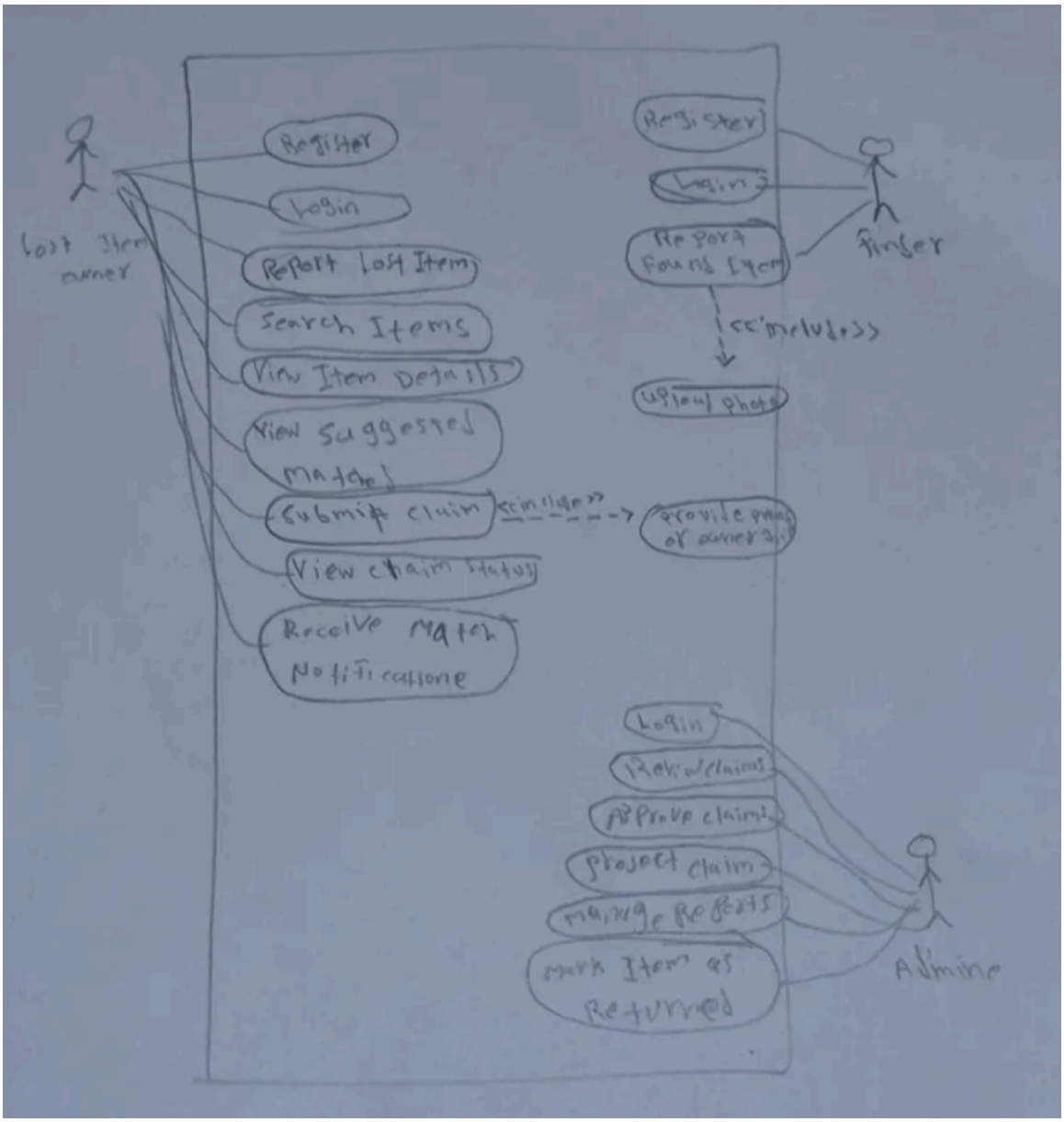
▼ Main System Functionality

- Register
- Login
- Report Lost Item
- Report Found Item
- Search Items
- View Suggested Matches
- Submit Claim
- Provide Proof of Ownership
- View Claim Status
- Receive Match Notification
- Review Claim
- Verify Ownership
- Manage Reports
- Mark Item as Returned

---

week 5 | 20 - 24 SEP

▼ Use case model (V1)



▼ User scenarios (V1)

1- As a student who lost an item, I want to receive a notification when a found item matches my lost-item report, so that I can quickly verify and recover my item.

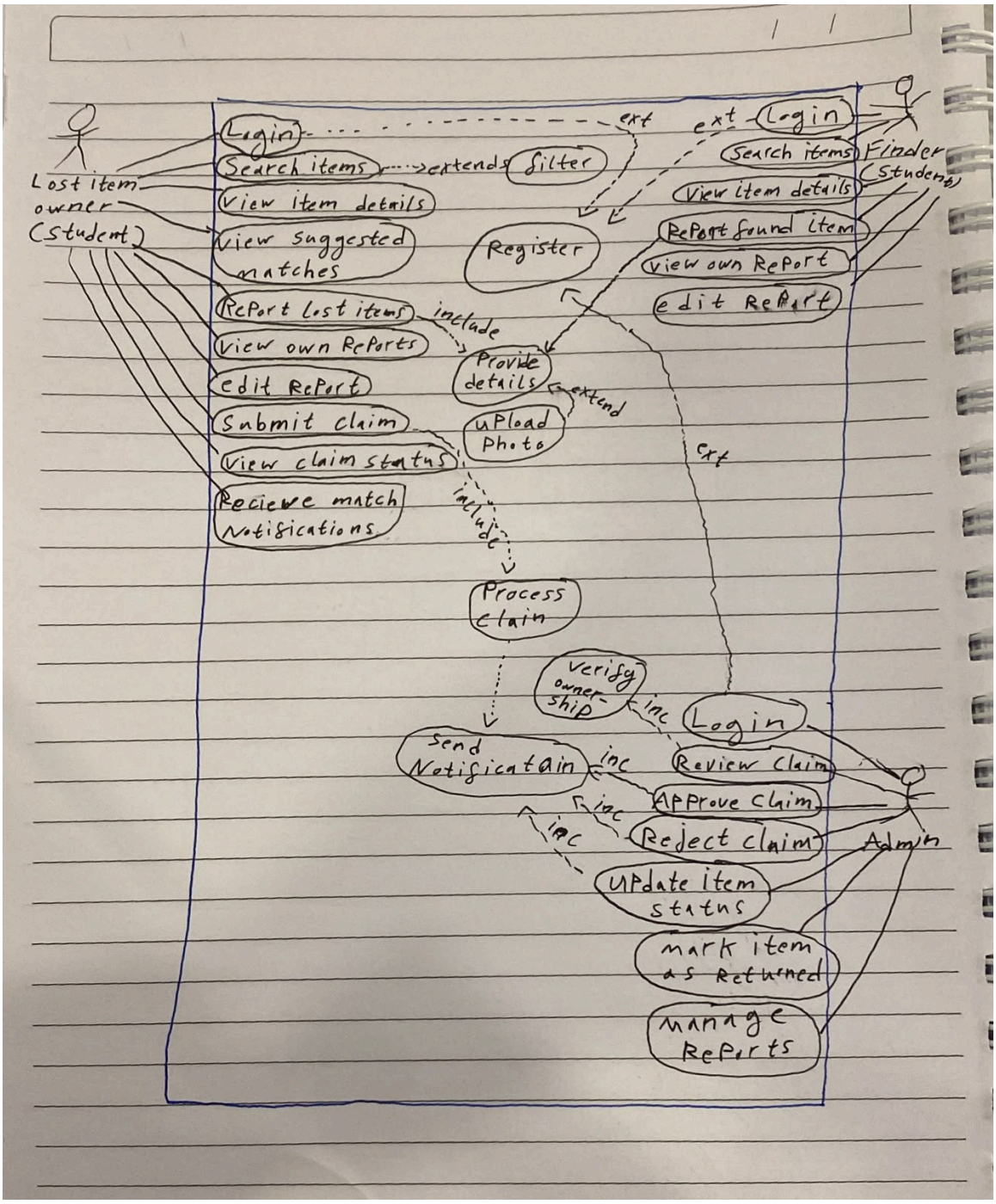
2- As a finder, I want to report a found item with a photo, description, and location, so that the rightful owner can identify and recover it.

3-As an administrator, I want to review ownership claims, so that only the rightful owner can receive the found item.

---

week 6 | 27-30 SEP:

▼ Use case model (V2)



▼ Core Use Cases Identification

1. Report Lost Item
2. Report Found Item
3. Search Items
4. View Suggested Matches
5. Receive Match Notification
6. Submit Claim
7. View Claim Status
8. Review Claim
9. Manage Reports
10. Mark Item as Returned

▼ Use Case Specification Development

1. Report Lost Item
   Name: Report Lost Item
   Actor: Lost Item Owner / Student
   Goal: Report a lost item so it can be searched for and matched with found items.
   Preconditions: Student is logged in.
   Summary: The student enters the lost-item information and submits the report.
   Related Use Cases:

- <<include>> Provide Item Details
- Upload Photo <<extend>> Report Lost Item
  Steps:
  Actor actions System responses

1. Student chooses Report Lost Item. 2. System displays the reporting form.
2. Student enters the item details. 4. System accepts the entered details.
3. Student may upload a photo. 6. System attaches the photo if provided.
4. Student submits the report. 8. System stores the report.
    a. System checks for possible matches.
    b. System confirms successful creation.

Postconditions: The lost-item report is stored and available for matching.
This follows the Chapter 4.6 structure of Name, Actors, Goals, Preconditions, Summary, Related Use Cases, Steps, and Postconditions.

2. Report Found Item
Name: Report Found Item
Actor: Finder / Student
Goal: Report a found item so its owner can be located.
Preconditions: Finder is logged in.
Summary: The finder enters information about the found item and submits it.
Related Use Cases:

- <<include>> Provide Item Details
- Upload Photo <<extend>> Report Found Item
  Steps:
  Actor actions System responses

1. Finder chooses Report Found Item. 2. System displays the reporting form.
2. Finder enters item details. 4. System accepts the details.
3. Finder may upload a photo. 6. System attaches the photo if provided.
4. Finder submits the report. 8. System stores the report.
    a. System compares it with lost-item reports.
    b. System records any possible matches.

Postconditions: The found-item report is stored and available for matching.

3. Submit Claim
Name: Submit Claim
Actor: Lost Item Owner / Student
Goal: Submit a claim for an item the student believes belongs to them.
Preconditions: Student is logged in and has identified a found item.
Summary: The student submits a claim and the system records it as Pending.
Related Use Cases: View Item Details, View Suggested Matches, View Claim Status

Steps:
Actor actions System responses

1. Student views a found item. 2. System displays its details.
2. Student chooses Submit Claim. 4. System asks for confirmation.
3. Student confirms. 6. System stores the claim.
    a. System changes its status to Pending.
    b. System makes it available for administrator review.

Postconditions: A Pending claim exists.

4. Review Claim
Name: Review Claim
Actor: Administrator
Goal: Decide whether a claim should be approved or rejected.
Preconditions: Administrator is logged in and a Pending claim exists.
Summary: The administrator reviews the claim and verifies whether the claimant is the rightful owner.
Related Use Cases:

- <<include>> Verify Ownership
- Approve Claim
- Reject Claim
  Steps:
  Actor actions System responses

1. Administrator opens pending claims. 2. System displays the pending claims.
2. Administrator selects a claim. 4. System displays the related lost and found reports.
3. Administrator verifies the information. 6. System allows approval or rejection.
4. Administrator chooses Approve or Reject. 8. System updates the claim status.
5. System sends a status notification.

▼ User story (V2)

Search Items
As a Lost Item Owner, I want to search reported found items, so that I can look for my lost item.

View Suggested Matches
As a Lost Item Owner, I want to view suggested matches, so that I can quickly check items that may be mine.

Receive Match Notification
As a Lost Item Owner, I want to receive a notification when a possible match is found, so that I can check it quickly.

View Claim Status
As a Lost Item Owner, I want to view my claim status, so that I know whether it is Pending, Approved, or Rejected.

Manage Reports
As an Administrator, I want to manage incorrect, duplicate, or suspicious reports, so that the system remains accurate.

Mark Item as Returned
As an Administrator, I want to mark an item as Returned, so that the related reports can be marked as Resolved.

---

week 7 | 3- 10 OCT :

Just meeting with D.naif

---

week 8 | 11- 18 OCT :

▼ Non-Functional Requirements (NFRs)

- **Privacy & Identity Protection:** The system must block direct personal communication between the finder and the owner, routing all updates through system notifications. *(Justification: Protects student contact info and prevents harassment).*
- **Automated Match Performance:** The matching engine must compare new reports and suggest potential matches within 3 seconds. *(Justification: Ensures real-time alerts as soon as a matching item is entered).*
- **Image Compression:** The system must compress uploaded images (max 3MB) while preserving the clarity of distinguishing marks. *(Justification: Maintains storage efficiency while allowing admins to verify visual proof).*
- **Role-Based Access Control:** The system must strictly separate student account privileges from administrator privileges. *(Justification: Ensures data integrity and prevents regular users from self-approving claims).*
- **Mobile-Responsive Usability:** The web application interface must be fully responsive for mobile phones. *(Justification: Students will most often use the system via smartphones immediately upon losing/finding an item on campus).*

▼ 3 Core Use Cases for BPMN Modeling

▼ Use Case 1: Recover a Lost Item (Submit & Review Claim)

Field	Content
Use case	**Submit & Review Ownership Claim**

Primary actor	Student (Lost Item Owner)
Supporting actor	Lost & Found Administrator
Trigger	A student identifies a potential match for their lost item and wishes to claim it
Main success scenario	1. Student views found item details and submits an ownership claim with photos/proof. 2. System sets claim status to "Pending" and routes it to the Administrator. 3. Administrator reviews the claim details and visual proof. 4. Administrator verifies that the student is the rightful owner. 5. Administrator approves the claim, hands over the item, and marks the status as "Returned".
Extensions	1a. Incomplete claim details: Administrator requests additional proof; student updates the claim. 4a. Ownership not confirmed: Administrator rejects the claim; system notifies the student and updates status to "Rejected".
Postcondition	Item is successfully returned to its owner and report is resolved, or claim is rejected and item remains in the system.

▼ Use Case 2: Report Found Item & Match Notification

Field	**Content**
Use case	**Report Found Item & Auto-Matching**
Primary actor	Student (Finder)
Supporting actor	System / Lost & Found Administrator
Trigger	A student finds an unattended item on campus
Main success scenario	1. Finder submits a found-item report with item details, location, and photo.

	2. Finder hands over the physical item to the Lost & Found office. 3. System saves the report and executes the automated matching algorithm against active lost-item reports. 4. System identifies a strong matching lost-item report. 5. System automatically generates and sends a match notification to the lost item owner.
Extensions	3a. No matching lost-item report found: System saves the found-item report in the active register; report stays open. 3b. Incomplete found report: System prompts finder to complete required fields before saving.
Postcondition	Found item is registered and potential owner is notified, or item remains open in register awaiting future matches.

▼ Use Case 3: Report Lost Item & Manual Search Filter

Field	**Content**
Use case	**Report Lost Item & Manual Search**
Primary actor	Student (Lost Item Owner)
Supporting actor	System
Trigger	A student realizes an item is missing on campus
Main success scenario	1. Student submits a lost-item report specifying category, building, date, and description. 2. System confirms report creation and performs an initial match check. 3. Student navigates to the search portal and applies filters (category, location, date).

	4. System displays matching found items. 5. Student locates their item in the results and proceeds to submit an ownership claim.
Extensions	2a. Direct match detected immediately: System prompts student to view suggested matches. 4a. Item not listed in search results: System keeps lost-item report active and enables automated alerts for future found items.
Postcondition	Lost item report is registered and student initiates a claim or receives automated alerts for future entries.

week 9 | 18- 24 OCT :

No new tasks for the course project

week 10&11 | 25OCT - 6 NOV :

▼ Use Case 4: Manage Suspicious or Duplicate Reports

Field	Content
Use case	**Manage Suspicious or Duplicate Reports**
Primary actor	Lost & Found Administrator
Supporting actor	System / Student
Trigger	Administrator reviews system logs or receives a report regarding inappropriate, duplicate, or suspicious listings
Main success scenario	1. Administrator accesses the dashboard and views active listings. 2. Administrator selects a suspicious or duplicate report. 3. Administrator takes action by updating or deleting the report. 4. System updates the report status to "Modified" or "Removed".

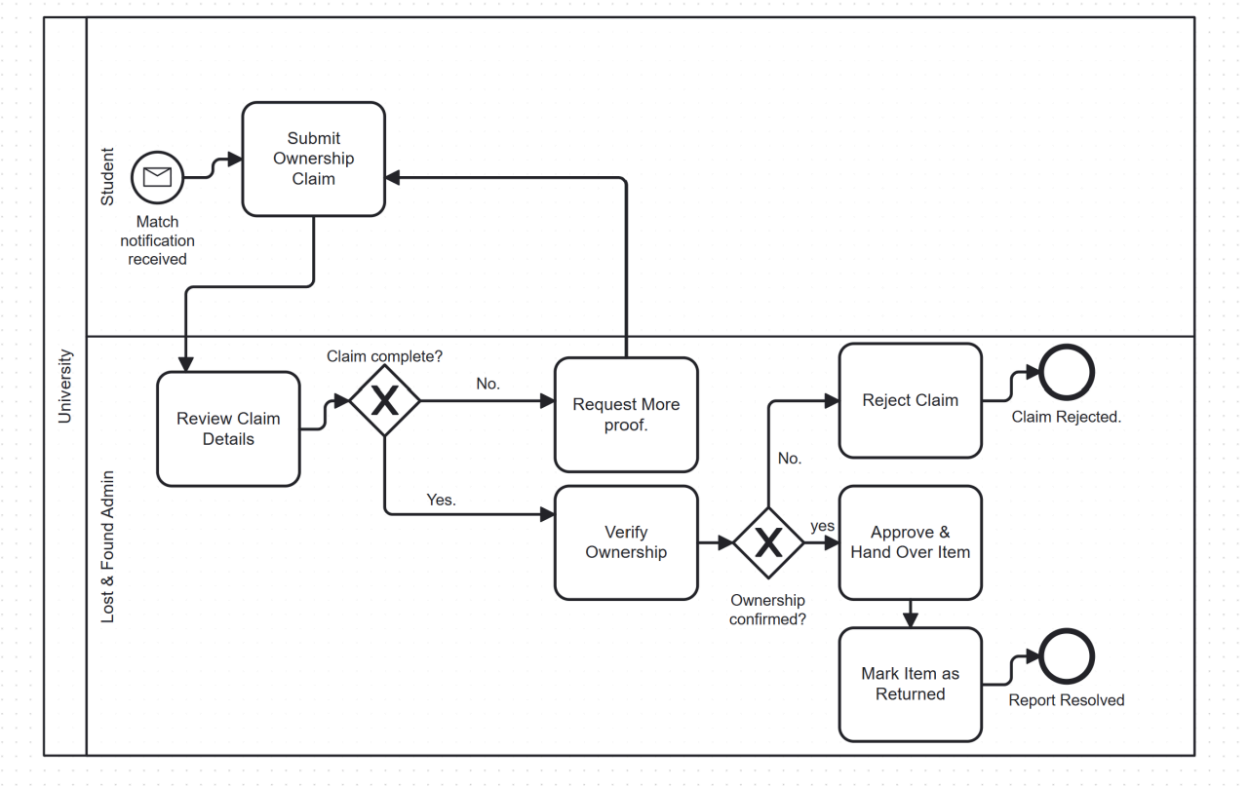
	5. System notifies the report author of the administrative action taken.
Extensions	3a. Report is valid: Administrator dismisses the review flag; report remains active. 3b. Report violates terms: Administrator suspends the user account.
Postcondition	Suspicious or duplicate report is managed, maintaining database integrity.

▼ Use Case 5: Mark Item as Returned & Resolve Case

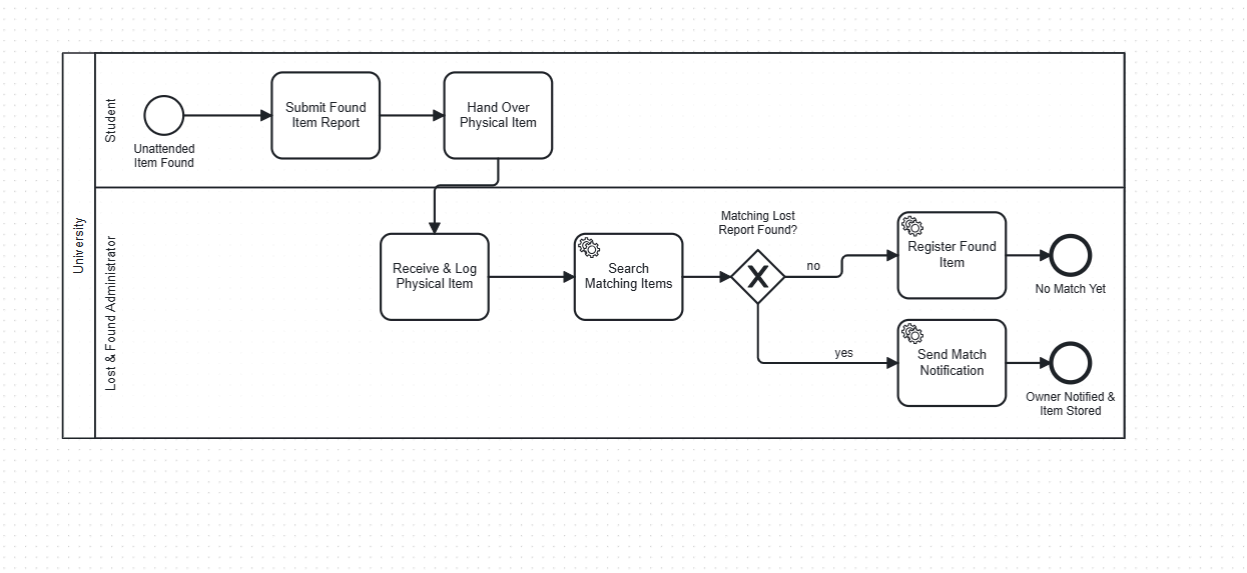
Field	Content
Use case	**Mark Item as Returned & Resolve Case**
Primary actor	Lost & Found Administrator
Supporting actor	Student (Lost Item Owner) / System
Trigger	An approved claimant arrives at the Lost & Found office to retrieve their item
Main success scenario	1. Administrator verifies the student's identity and presents the physical item. 2. Student confirms receipt of the item. 3. Administrator opens the approved claim record in the system. 4. Administrator clicks "Mark as Returned". 5. System updates the item status to "Returned" and marks the associated lost/found reports as "Resolved".
Extensions	1a. Student identity does not match approved claim: Administrator holds the item; status remains pending verification.
Postcondition	Item is physically handed over, item status is "Returned", and the case is closed as "Resolved".

▼ Draw BPMN

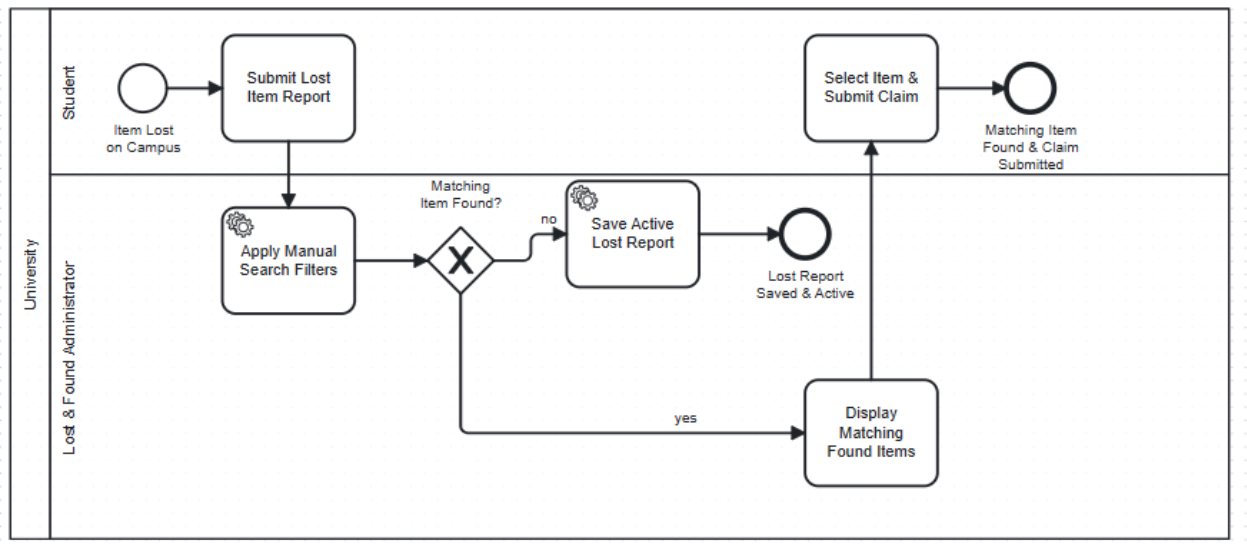
Use Case 1: Recover a Lost Item (Submit & Review Claim)



Use Case 2: Report Found Item & Match Notification



Use Case 3: Report Lost Item & Manual Search Filter



Use Case 4: Manage Suspicious or Duplicate Reports

(حط الصوره هنا يا يزن)

Use Case 5: Mark Item as Returned & Resolve Case

(حط الصوره هنا يا يزن)